MATHEMATICS

METHOD OF DIFFERENTIATION THEORY SHEET ENGLISH

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1.1 DIFFERENTIATION BY FIRST PRINCIPLE

Let $y = f(x)$; $y + \Delta y = f(x + \Delta x)$

$$\therefore \quad \frac{\Delta y}{\Delta x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$$ (average rate of change of function)

$$\therefore \quad \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \quad \text{....(1)}$$

(1) denotes the instantaneous rate of change of function.

It is also known as :

1. Differentiation by fundamental definition of calculus.
2. Differentiation by delta method.
3. Differentiation by ab - initio method.

Note that if $y = f(x)$ then the symbols $\frac{dy}{dx} = Dy = f'(x) = y_1$ or y' have the same meaning.

However a dot, denotes the time derivative. e.g. $\dot{S} = \frac{dS}{dt}$; $\dot{\theta} = \frac{d\theta}{dt}$ etc.

Q.1 Find derivative of function $f(x) = x^2$ by first principle with respect to x.

Ans. $2x$

Sol. : $f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h} = 2x.$

Q.2 Find derivative of function $f(x) = \tan x$ by first principle with respect to x.

Ans. $\sec^2 x$

Sol. : $f'(x) = \lim_{h \to 0} \frac{\tan(x+h) - \tan x}{h} = \lim_{h \to 0} \frac{\frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x}}{h}$

$$= \lim_{h \to 0} \frac{\sin(x+h-x)}{h \cos x . \cos(x+h)} = \sec^2 x.$$

Q.3 Find derivative of function $f(x) = e^{\sin x}$ by first principle with respect to x.

Ans $e^{\sin x} \cos x$

Sol. $f'(x) = \lim_{h \to 0} \frac{e^{\sin(x+h)} - e^{\sin x}}{h} = \lim_{h \to 0} e^{\sin x} \frac{\left[e^{\sin(x+h) - \sin x} - 1\right]}{\sin(x+h) - \sin x} \left(\frac{\sin(x+h) - \sin x}{h}\right)$

$$= e^{\sin x} \lim_{h \to 0} \frac{\sin(x+h) - \sin x}{h} = e^{\sin x} \lim_{h \to 0} \frac{2\cos\left(\frac{x+h+x}{2}\right)\sin\left(\frac{x+h-x}{2}\right)}{h}$$

$$= e^{\sin x} \lim_{h \to 0} \left\{ \cos\left(x + \frac{h}{2}\right) \frac{\sin(h/2)}{h/2} \right\} = e^{\sin x} \cos x$$

1.2 DERIVATIVES OF STANDARD FUNCTIONS

(i) $\frac{d}{dx}(\text{constant}) = 0$ (ii) $\frac{d}{dx}(ax) = a$ (iii) $\frac{d}{dx}(x^n) = nx^{n-1}$

(iv) $\frac{d}{dx} e^x = e^x$ (v) $\frac{d}{dx}(a^x) = a^x \log_e a$ (vi) $\frac{d}{dx}(\log_e x) = 1/x$

(vii) $\frac{d}{dx}(\log_a x) = \frac{1}{x \log_e a}$ (viii) $\frac{d}{dx}(\sin x) = \cos x$ (ix) $\frac{d}{dx}(\cos x) = -\sin x$

(x) $\frac{d}{dx}(\tan x) = \sec^2 x$ (xi) $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$ (xii) $\frac{d}{dx}(\sec x) = \sec x \tan x$

(xiii) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$ (xiv) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$

(xv) $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$ (xvi) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

(xvii) $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$ (xviii) $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}; |x| > 1$

(xix) $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}; |x| > 1$

Find f'(x)

1. $f(x) = \sqrt{x}$

Ans. $\frac{1}{2\sqrt{x}}$

Sol. $\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$

2. $f(x) = \frac{1}{x}$

Ans. $\frac{-1}{x^2}$

Sol. $\frac{d}{dx}\frac{1}{x} = -\frac{1}{x^2}$

3. $f(x) = \frac{1}{x^7}$

Ans. $\frac{-7}{x^8}$

Sol. $\frac{d}{dx}\frac{1}{x^7} = \frac{d}{dx}x^{-7} = -7x^{-8}$

4. $f(x) = \frac{1}{\sin x}$

Ans. $-\text{cosec } x \cot x$

Sol. $\frac{d}{dx}\text{cosec } x = -\text{cosec } x \cot x$

5. $f(x) = \tan(\tan^{-1} x)$

Ans. 1

Sol. $\frac{d}{dx}\tan\left(\tan^{-1} x\right) = \frac{d}{dx}(x) = 1$

6. $f(x) = \frac{1}{\log_x e}$

Ans. $\frac{1}{x}$

Sol. $\frac{1}{dx}\frac{1}{\log_x e} = \frac{d}{dx}\log_e{}^x = \frac{1}{x}$

7. $f(x) = e^{\ln(\cot^{-1}x)}$

Ans. $\frac{-1}{1+x^2}$

Sol. $\frac{d}{dx}e^{\ell n\left(\cot^{-1}x\right)} = \frac{d}{dx}\cot^{-1} x = -\frac{1}{1+x^2}$

8. $f(x) = a^{\pi}$

Ans. 0

Sol. $\frac{d}{dx}a^{\pi} = 0$ (any constant)

9. $f(x) = (\cos^{-1} x + \sin^{-1} x)^2$

Ans. 0

Sol. $\frac{d}{dx}\left(\sin^{-1}+\cos^{-1} x\right)^2 = \frac{d}{dx}\frac{\pi^2}{4} = 0$ (ITF property)

10. $f(x) = \cos^4\left(\frac{x}{2}\right) - \sin^4\left(\frac{x}{2}\right)$

Ans. $-\sin x$

Sol. $\frac{d}{dx}\left(\cos^4\left(\frac{x}{2}\right)-\sin^4\left(\frac{x}{2}\right)\right)$

$= \frac{d}{dx}\left(\cos^2\frac{x}{2}+\sin^2\frac{x}{2}\right)\left(\cos^2\frac{x}{2}-\sin^2\frac{x}{2}\right)$

$\frac{d}{dx}\cos x = -\sin x$

11. $f(x) = \frac{x^3+x^4}{1+x}$

Ans. $3x^2$

Sol. $\frac{d}{dx}\frac{x^3+x^4}{1+x} = \frac{d}{dx}x^3\frac{(1+x)}{(1+x)} = 3x^2$

12. $f(x) = 2^x e^x$

Ans. $(2e)^x \ln(2e)$

Sol. $\frac{d}{dx}2^x e^x = \frac{d}{dx}(2e)^x = (2e)^x \cdot \ell n(2e)$

13. $f(x) = 2^{\ln x}$

Ans. $(\ln 2)x^{\ln(2)-1}$

Sol. $\frac{d}{dx}2^{\ell nx}$

$= \frac{d}{dx}x^{\ell n 2}$

$= \ell n2 \cdot x^{(\ell n 2-1)}$

14. $f(x)=\dfrac{e^{3x}+e^{x}}{e^{2x}+1}$

Ans. e^{x}

Sol. $\dfrac{d}{dx}\dfrac{e^{3x}+e^{x}}{e^{2x}+1}=\dfrac{d}{dx}e^{x}\left(\dfrac{2^{2x}+1}{2^{2x}+1}\right)$

$=\dfrac{d}{dx}e^{x}=e^{x}$

1.3 IMPORTANT THEOREMS

There are certain basic theorems in differentiation:

1. $\dfrac{d}{dx}\,(f\pm g)=f'(x)\pm g'(x)$

2. $\dfrac{d}{dx}\,(k\,f(x))=k\,\dfrac{d}{dx}\,f(x)$

3. Product Rule

$\dfrac{d}{dx}\,(f(x)\,.\,g(x))=f(x)\,g'(x)+g(x)\,f'(x)$

4. Quotient Rule

$\dfrac{d}{dx}\left(\dfrac{f(x)}{g(x)}\right)=\dfrac{g(x)\,f'(x)-f(x)g'(x)}{g^{2}(x)}$

Q.1 Find $\dfrac{dy}{dx}$

(i) $y=x\sin^{-1}x$

Ans. $\sin^{-1}(x)+\dfrac{x}{\sqrt{1-x^{2}}}$

Sol. $\dfrac{d}{dx}\left(x\sin^{-1}x\right)=x\cdot\dfrac{1}{\sqrt{1-x^{2}}}+\sin^{-1}x$

(ii) $y=e^{x}\tan^{-1}(x)$

Ans. $e^{x}\tan^{-1}(x)+\dfrac{e^{x}}{1+x^{2}}$

Sol. $\frac{d}{dx}e^x\left(\tan^{-1}x\right)=e^x\left(\frac{1}{1+x^2}\right)+\left(\tan^{-1}x\right)\left(e^x\right)$

$=e^x\left[\frac{1}{1+x^2}+\tan^{-1}x\right]$

(iii) $y=x\ln x$

Ans. $1+\ln x$

Sol. $\frac{d}{dx}\left(x\ell nx\right)=x\cdot\frac{1}{x}+\ell nx=\left(1+\ell nx\right)$

(iv) $y=\frac{x}{\sin x}$

Ans. $\frac{\sin x-x\cos x}{\sin^2 x}$

Sol. $\frac{d}{dx}\frac{x}{\sin x}=\frac{\sin x\left(1\right)-x\left(\cos x\right)}{\left(\sin^2 x\right)}=\frac{\sin x-x\cos x}{\sin^2 x}$

(v) $y=\frac{1-\ln x}{1+\ln x}$

Ans. $\frac{-2}{x\left(1+\ln x\right)^2}$

Sol. $\frac{d}{dx}\left(\frac{1-\ell nx}{1+\ell nx}\right)=\frac{\left(1+\ell nx\right)\left(-\frac{1}{x}\right)-\left(1-\ell nx\right)\left(\frac{1}{x}\right)}{\left(1+\ell nx\right)^2}$

$=\frac{-\left[1+\ell nx+1-\ell nx\right]}{x\left(1+\ell nx\right)^2}$

$=\frac{-2}{x\left(1+\ell nx\right)^2}$

(vi) $y=\frac{x^3+x^2}{e^x}$

Ans. $\frac{-x^3+2x^2+2x}{e^x}$

Sol. $\frac{d}{dx}\frac{x^3+x^2}{e^x}=\frac{e^x\left(3x^2+2x\right)-e^x\left(x^3+x^2\right)}{\left(e^x\right)^2}$

$$=\frac{e^x\left[3x^2+2x-x^3-x^2\right]}{\left(e^x\right)^2}$$

$$=\frac{e^x\left(2x^2-x^3+2x\right)}{\left(e^x\right)^2}$$

Q.2 Prove product rule with the help of first principle.

Sol. $\frac{d}{dx}f.g=\lim_{h\to0}\frac{f\left(x+h\right)g\left(x+h\right)-f\left(x\right)g\left(x\right)}{h}$

$$\lim_{h\to0}\frac{\left\{f\left(x+h\right)-f\left(x\right)\right\}g\left(x+h\right)+f\left(x\right)g\left(x+h\right)-f\left(x\right)g\left(x\right)}{h}$$

$$\lim_{h\to0}\frac{\left\{f\left(x+h\right)-f\left(x\right)\right\}g\left(x+h\right)}{h}+\lim_{h\to0}f\left(x\right)\frac{\left\{g\left(x+b\right)-9\left(x\right)\right\}}{h}$$

$$f'\left(x\right)g\left(x\right)+g'\left(x\right)f\left(x\right)$$

Q.3 Prove that (fgh)' = ((fg)' h + (gh)' f + (hf)' g)/2

Sol. (fgh)' = f'gh + fg'h + fgh'

$$=\frac{2f'gh+2fg'h2fgh'}{2}$$

$$=\frac{f'gh+f'gh+fg'h+fg'h+fgh'+fgh'}{2}$$

$$=\frac{\left(fg\right)'h+\left(gh\right)'f+\left(hf\right)'g}{2}$$

Q.4 f(x) = (1 + x) $\sqrt{3+x^2}$ $\sqrt{9+x^3}$ find f'(– 1)

Ans. $4\sqrt{2}$

Sol. $f'(x)=\sqrt{3+x^2}\sqrt{9+x^3}+(1+x)\frac{2x}{2\sqrt{3+x^2}}\sqrt{9+x^3}+(1+x)\left(\sqrt{3+x^2}\right)\frac{3x^2}{2\sqrt{9+x^2}}$

$f'(-1)=2\left(2\sqrt{2}\right)=4\sqrt{2}$

Q.5 $f(x) = 1 + x + x^2 + \ldots\ldots\ldots\ldots + x^{100}$ find $f'(1)$

Ans. 5050

Sol. $f'(x) = 1 + 2x + 3x^2 + \ldots\ldots + 100x^{99}$

$f'(1)=\sum_{n=1}^{100} n = 5050$

Q.6 Find $\frac{dy}{dx}$

(i) $y=\frac{x^4+x^2+1}{x^2+x+1}$

Ans. (i) $2x - 1$

Sol. $\frac{d}{dx}\left(\frac{x^4+x^2+1}{x^2+x+1}\right)=\frac{d}{dx}\frac{\left(x^2-x+1\right)\left(x^2+x+1\right)}{\left(x^2+x+1\right)}$

$=\frac{d}{dx}\left(x^2-x+1\right)$

$= 2x - 1$

(ii) $y=\frac{\tan^2 x.\sin^2 x}{\tan^2 x-\sin^2 x}$

Ans. 0

Sol. $\frac{d}{dx}\frac{\tan^2 x\cdot\sin^2 x}{\tan^2 x-\sin^2 x}=\frac{d}{dx}\frac{\tan^2 x\cdot\sin^2 x}{\sin^2 x\left[\frac{1}{\cos^2 x}-1\right]}$

$=\frac{d}{dx}\frac{\tan^2 x\cdot\sin^2 x}{\frac{\sin^2 x}{\cos^2 x}\left(1-\cos^2 x\right)}$

$$= \frac{d}{dx}\frac{\sin^2 x}{\sin^2 x} = 0$$

(iii) $y = \dfrac{\sec x + \tan x - 1}{-\sec x + \tan x + 1}$

Ans. $\sec x (\sec x + \tan x)$

Sol. $\dfrac{d}{dx}\dfrac{\sec x + \tan x - 1}{\tan x - \sec x + 1} = \dfrac{d}{dx}\dfrac{(\sec x + \tan x) - (\sec^2 x - \tan^2 x)}{\tan x - \sec x + 1}$

$1 = \sec^2 x - \tan^2 x = \dfrac{d}{dx}\dfrac{(\sec x + \tan x)[1 - \sec x + \tan x]}{(\tan x - \sec x + 1)}$

$= \sec x \tan x + \sec^2 x$

$= \sec x(\tan x + \sec x)$

(iv) $y = \dfrac{\tan^{-1}(x) - \cot^{-1}(x)}{\tan^{-1}(x) + \cot^{-1}(x)}$

Ans. $\dfrac{4}{\pi(1 + x^2)}$

Sol. $\dfrac{d}{dx}\dfrac{\tan^{-1} x - \cot^{-1} x}{\tan^{-1} x + \cot^{-1} x} = \dfrac{2}{\pi}\left(\dfrac{d}{dx}\tan^{-1} x - \dfrac{d}{dx}\cot^{-1} x\right)$

$\therefore \tan^{-1} x + \cot^{-1} x = \dfrac{\pi}{2} = \dfrac{2}{\pi}\left(\dfrac{1}{1 + x^2} - \left(-\dfrac{1}{1 + x^2}\right)\right)$

$= \dfrac{2}{\pi}\cdot\dfrac{2}{(1 + x^2)} = \dfrac{4}{\pi(1 + x^2)}$

(v) $y = \dfrac{x^3 - 3x + 2}{x - 1}$

Ans. $2x + 1$

Sol. $\dfrac{d}{dx}\left(\dfrac{x^3 - 3x + 2}{x - 1}\right) = \dfrac{d}{dx}\dfrac{(x - 1)(x^2 + x - 2)}{(x - 1)}$

$= 2x + 1$

(vi) $y = \dfrac{x^3 + x^2 + x}{x^2 + 1}$

Ans. $1 + \dfrac{2x}{(1+x^2)^2}$

Sol. $y = \dfrac{x(1+x^2) + x^2 + 1 - 1}{(1+x^2)} = x + 1 - \dfrac{1}{(1+x^2)}$

$y' = 1 + 0 - \dfrac{(1+x^2)\cdot 0 - 1\cdot 2x}{(1+x^2)^2} = 1 + \dfrac{2x}{(1+x^2)^2}$

Q.7 F(x) = f(x) . g(x). h(x) $\forall$ x $\in$ R at $x = x_0$ $F'(x_0) = 21\, F(x_0)$, $f'(x_0) = 4f(x_0)$, $g'(x_0) = 7\, g(x_0)$, $h'(x_0) = k\, h(x_0)$ find k.

Ans. k = 10

Sol. F(x) = f(x) . g(x).h(x) $\forall$ x $\in$ R

$F'(x)f'(x)g(x)h(x)$

$F'(x) = f'(x)g(x)h(x) + F(x)g'(x)h(x) + F(x)g(x)h'(x)$

$F'(x_0) = 4F(x_0)g'(x_0)h(x_0) + f(x_0)7g(x_0)h(x_0) + F(x_0)g(x_0)kh(x_0)$

$F'(x_0) = 11F(x_0)g(x_0)(x_0) + kf(x_0)g(x_0)h(x_0)$

$F'(x_0) = (11+k)f(x)g(x_0)h(x_0)$

$21f(x_0) = (11+k) + (x_0)$

11 + k = 21

k = 10

Q.8 f, g, h are differentiable function f(0) = 1, g (0) = 2, h(0) = 3, (fg)' (0) = 6, (gh)' (0) = 4 (hf)' (0)= 5 find (fgh)' (0)

Ans 16

Sol. $(fgh)'(0) = f\,g\,h' + g\,h\,f' + h\,f\,g' = \dfrac{(fg)'h + (gh)'f + (hf)'g}{2}$ **(note)**

$\therefore \quad (fgh)'(0) = \dfrac{(fg)'(0)\cdot h(0) + (gh)'(0)\cdot f(0) + (hf)'(0)\cdot g(0)}{2}$

$= \dfrac{6\cdot 3 + 4\cdot 1 + 5\cdot 2}{2} = 16$

5. Chain Rule

$$\frac{d}{dx}(f(g(x))) = f'(g(x))\,g'(x)$$

This rule is called the chain rule of differentiation and can be written as

$$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{dz}{dx}$$

Note that an important inference obtained from the chain rule is that

$$\frac{dy}{dy} = 1 = \frac{dy}{dx}\cdot\frac{dx}{dy} \quad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{dx/dy}$$

Q.9 Find $\frac{dy}{dx}$

(i) $y = f(g(x))$

Ans. $f'(g(x))\,g'(x)$

(ii) $y = e^{x^2}$

Ans. $2xe^{x^2}$

Sol. $\frac{d}{dx}e^{x^2} = e^{x^2}\left(\frac{d}{dx}x^2\right) = e^{x^2}(2x)$

(iii) $y = \frac{1}{f(x^3)}$

Ans. $\frac{-1}{\left(f\left(x^3\right)\right)^2}f'\left(x^3\right)3x^2$

Sol. $\frac{d}{dx}\frac{1}{f\left(x^3\right)} = \frac{-1}{\left(f\left(x^3\right)\right)^2}f'\left(x^3\right)\cdot\left(3x^2\right)$

(iv) $y = f(g(h(x^2)))$

Ans. $f'(g(h(x^2)))\,g'(h(x^2))h'(x^2)(2x)$

Sol. $\frac{d}{dx}f\left(g\left(h\left(x^2\right)\right)\right) = f'(g(h(x^2)))\cdot g'(h(x^2))\cdot h'(x^2)\,(2x)$

(v) $y = f\left(\frac{1}{x}\right)$

Ans. $f'\left(\frac{1}{x}\right)\left(\frac{-1}{x^2}\right)$

Sol. $\frac{d}{dx}f\left(\frac{1}{x}\right)=f'\left(\frac{1}{x}\right)\cdot\left(\frac{-1}{x^2}\right)$

(vi) $y=(f(x))^4$

Ans. $4(f(x))^3 f'(x)$

Sol. $y=\left(f\left(x\right)\right)^4=\frac{dy}{dx}=4(f(x))^3\cdot f'(x)$

(vii) $f(x)=\sin^3 x$

Ans. $3\sin^2(x)\cos x$

Sol. $f\left(x\right)=\sin^3 x$

$\frac{dy}{dx}=3\sin^2 x\cdot\cos x$

(viii) $y=\cos^4\sqrt{x}$

Ans. $4\cos^3\left(\sqrt{x}\right)\left(-\sin\sqrt{x}\right)\left(\frac{1}{2\sqrt{x}}\right)$

Sol. $\frac{d}{dx}\cos^4\sqrt{x}=4\cos^3\sqrt{x}\left(-\sin\sqrt{x}\right)\cdot\frac{1}{2\sqrt{x}}$

(ix) $y=\sqrt{f(x)}+f(\sqrt{x})$

Ans. $\frac{f'(x)}{2\sqrt{f(x)}}+f'\left(\sqrt{x}\right)\left(\frac{1}{2\sqrt{x}}\right)$

Sol. $\frac{d}{dx}\left(\sqrt{f(x)}+f\left(\sqrt{x}\right)\right)=\frac{f'(x)}{2\sqrt{f(x)}}+f'\left(\sqrt{x}\right)\frac{1}{2\sqrt{x}}$

(x) $y=\ln^3(\tan^2(x^4))$

Ans. $3\ln^2\left(\tan^2\left(x^4\right)\right)\left(\frac{1}{\tan^2\left(x^4\right)}\right)\left(2\tan\left(x^4\right)\right)\left(\sec^2\left(x^4\right)\right)\left(4x^3\right)$

Sol. $\frac{d}{dx}\ell n^3\left(\tan^2\left(x^4\right)\right)$

$$=3\cdot\ell n^2\left(\tan^2\left(x^4\right)\right)\cdot\frac{1}{\tan^2\left(x^4\right)}\cdot 2\tan\left(x^4\right)\cdot\sec^2\left(x^4\right)\left(4x^3\right)$$

(xi) $y = fog(x) + gof(x)$

Ans. $f'(g(x))g'(x) + g'f(x)\, f'(x)$

Sol. $\frac{d}{dx}fog(x) + \frac{d}{dx}gof(x)$

$= f'(g(x))\,(g'(x)) + g'(f(x))\cdot f'(x)$

(xii) $y = \exp\left(\cos^3\left(\tan^{-1}\left(x^2\right)\right)\right)$

Ans. $\exp\left(\cos^3\left(\tan^{-1}\left(x^2\right)\right)\right)\left(3\cos^2\left(\tan^{-1}\left(x^2\right)\right)\right)\left(-\sin\left(\tan^{-1}\left(x^2\right)\right)\right)\times\frac{2x}{1+x^4}$

Sol. $\frac{d}{dx}e^{\cos^3\left(\tan^{-1}\left(x^2\right)\right)}$

$$=e^{\cos^3\left(\tan^{-1}\left(x^2\right)\right)}\cdot 3\cos^2\left(\tan^{-1}\left(x^2\right)\right)\left(-\sin\left(\tan^{-1}\left(x^2\right)\right)\right)\frac{1}{1+x^4}(2x)$$

(xiii) $y = \sec^2\left((f(x))^3\right)$

Ans. $2\sec^2\left(\left(f(x)\right)^3\right)\tan\left(f(x)^3\right)3\left(f(x)\right)^2 f'(x)$

Sol. $\frac{d}{dx}\sec^2\left(\left(f(x)\right)^3\right)$

$= 2\sec\left((f(x))^3\right)\cdot\sec(f(x)^3)\cdot\tan(f(x)^3)\cdot 3(f(x))^2\cdot f'(x)$

(xiv) $y = e^{ax}\sin bx$

Ans. $\sqrt{a^2+b^2}\,e^{ax}\sin(bx+\theta);\qquad \theta = \tan^{-1}\left(\frac{b}{a}\right)$

Sol. $\frac{d}{dx}\left(e^{ax}\cdot\sin bx\right)$

$= e^{ax}\cdot(a)\cdot\sin bx + e^{ax}\cos bx\cdot(b)$

$e^{ax}[a\cdot\sin bx + b\cdot\cos bx]$

$$e^{ax}\sqrt{a^2+b^2}\left[\frac{a}{\sqrt{a^2+b^2}}\sin bx+\frac{b}{\sqrt{a^2+b^2}}\cos bx\right]$$

Let $\cos\phi=\frac{a}{\sqrt{a^2+b^2}},\sin\phi=\frac{b}{\sqrt{a^2+b^2}}$

$\sqrt{a^2+b^2}\ (e^{ax})\sin bx\cos\phi+\cos bx\sin\phi$

$\sqrt{a^2+b^2}\ (e^{ax})\cdot\sin(bx+\phi)$ $\qquad\left\{\phi=\tan^{-1}\frac{b}{a}\right\}$

$\sqrt{a^2+b^2}\left(e^{ax}\right)\cdot\sin\left(bx+\tan^{-1}\frac{b}{a}\right)$

2. TECHNIQUES OF DIFFERENTIATION

2.1 LOGARITHMIC DIFFERENTIATION

The process of taking logarithm of the function first and then differentiate is called **Logarithmic differentiation**. It is often useful in following cases

Case-1 A function is the product or quotient of a number of functions

Q.1 If $y=\frac{x^{1/2}(1-2x)^{2/3}}{(2-3x)^{3/4}(3-4x)^{4/5}}$, then find $\frac{dy}{dx}$.

Ans. $y\left(\frac{1}{2x}-\frac{4}{3(1-2x)}+\frac{9}{4(2-3x)}+\frac{16}{5(3-4x)}\right)$

Sol. $\because\ y=\frac{x^{1/2}(1-2x)^{2/3}}{(2-3x)^{3/4}(3-4x)^{4/5}}$

taking $\log_e$ on both side

$$\ell n\,y=\frac{1}{2}\ \ell n\,x+\frac{2}{3}\ \ell n\,(1-2x)-\frac{3}{4}\ \ell n\,(2-3x)-\frac{4}{5}\ \ell n\,(3-4x)$$

$$\Rightarrow\quad \frac{1}{y}\frac{dy}{dx}=\frac{1}{2x}-\frac{4}{3(1-2x)}+\frac{9}{4(2-3x)}+\frac{16}{5(3-4x)}$$

$$\Rightarrow\quad \frac{dy}{dx}=y\left(\frac{1}{2x}-\frac{4}{3(1-2x)}+\frac{9}{4(2-3x)}+\frac{16}{5(3-4x)}\right)$$

Case-2 A function is of the form $[f(x)]^{g(x)}$ where f & g are both derivable,

Q.2 Find $\frac{d}{dx}\ x^{\sin x}$.

Ans. $x^{\sin x}\left[\frac{\sin x}{x}+\cos x.\log x\right]$

Sol. Let $y = x^{\sin x}$

$\log y = \sin x.\ \log x$

$\frac{1}{y}\frac{dy}{dx}=\frac{\sin x}{x}+\cos x\ .\ \log x$

$\therefore \frac{dy}{dx}= x^{\sin x}\left[\frac{\sin x}{x}+\cos x.\log x\right]$

AnotherMethod :

We can directly write the derivation of $[f(x)^{g(x)}]$ as given above.

$\frac{d}{dx}(x^{\sin x}) = x^{\sin x}\left[\frac{d}{dx}\sin x.\ \log x\right]$

$=x^{\sin x}\left[\frac{\sin x}{x}+\log x\cos x\right]$

Q.3 If $f(x)=(x+1)(x+2)(x+3).....(x+n)$ then $f'(0)$ is

Ans. $n!\left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+....+\frac{1}{n}\right)$

Sol. $f(x)=(x+1)\ (x+2)\ (x+3)\ .....\ (x+n)$

$\ell n(f(x))=\ell n(x+1)+\ell n(x+2)+.....+\ell n(x+n)$

$\frac{f'(x)}{f(x)}=\frac{1}{x+1}+\frac{1}{x+2}+.....+\frac{1}{x+n}$

$\therefore f'(0)=(n)!\left(1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n}\right)$

Q.4 $y=\sin x.\ \sin 2x.\ \sin 3x.\ \sin 4x$ find $\frac{dy}{dx}$

Ans. $y\ (\cot x+\cot 2x+3\cot 3x+4\cot 4x)$

Sol. $\ell ny=\ell n\sin x+\ell n\sin 2x+\ell n\sin 3x+\ell n\sin 4x$

$\frac{1}{y}\frac{dy}{dx}=\cot x+2\cot 2x+3\cot 3x+4\cot 4x$

Q.5 $f(x)=\frac{(1+x)(1+x^2)(1+x^3)}{(1+x^4)(1+x^5)(1+x^6)}$ find $f'(0)$

Ans. 1

Sol. $\ell ny = \ell n(1 + x) + \ell n(1 + x^2) + \ell n(1 + x^3) - \ell n(1 + x^4) - \ell n(1 + x^5) - \ell n(1 + x^6)$

$$\frac{dy}{dx} = y\left[\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{3x^2}{1+x^3} - \frac{4x^3}{1+x^4} - \frac{5x^4}{1+x^5} - \frac{6x^5}{1+x^6}\right]$$

$$\text{At } x = 0 \Rightarrow y = 1,\ \left.\frac{dy}{dx}\right|_{x=0} = 1$$

Q.6 Find $\frac{dy}{dx}$

(i) $y = (\sin x)^{\sin x}$

Ans. $(\sin x)^{\sin x}(\cos x)(1 + \ln \sin x)$

Sol. $$\frac{d}{dx}(\sin x)^{\sin x} = (\sin x)^{\sin x}\left[\cos x \cdot \ell n \sin x + \sin x \cdot \frac{\cos x}{\sin x}\right]$$

(ii) $y = x^{x^x}$

Ans. $$x^{x^x}\left(\frac{x^x}{x} + (\ln x)\, x^x (1 + \ln x)\right)$$

Sol. $$\frac{d}{dx} x^{x^x} = \ell ny = x^x \cdot \ell nx$$

$$\frac{1}{y}\frac{dy}{dx} = \left(x^x (1 + \ell nx)\, \ell nx + \frac{x^x}{x^x}\right)$$

$$x^{x^x}\left(\frac{x^x}{x} + (\ln x)\, x^x (1 + \ln x)\right)$$

(iii) $y = x^{\sin x}$

Ans. $$x^{\sin x}\left(\cos x \ln x + \frac{\sin x}{x}\right)$$

Sol. $$\frac{d}{dx} x^{\sin x}$$

$$= x^{\sin x}\left[\cos x \ell nx + \frac{\sin x}{x}\right]$$

(iv) $y = (\ln x)^x$

Ans. $(\ln x)^x\left(\ln(\ln x) + \frac{1}{\ln x}\right)$

Sol. $\frac{d}{dx}(\ell nx)^x = (\ell nx)^x\left[1 \cdot \ell n(\ell nx) + x \cdot \frac{1}{\ell nx} \times \frac{1}{x}\right]$

(v) $y = x^{\ln x}$

Ans. $\left((x)^{\ln x}\left(\frac{2\ln x}{x}\right)\right)$

Sol. $\frac{d}{dx}\left(x^{\ell nx}\right) = y = x^{\ell nx}$

$\ell ny = (\ell nx)^2$

$\frac{1}{y}\frac{dy}{dx} = \frac{2\ell nx}{x}$

$\frac{dy}{dx} = y\left(\frac{2\ell nx}{x}\right)$

$\left((x)^{\ln x}\left(\frac{2\ln x}{x}\right)\right)$

(vi) $y = x^x + x^{\sin x}$

Ans. $x^x(1+\ln x) + x^{\sin x}\left(\frac{\sin x}{x} + \cos x \ln x\right)$

Sol. Let $y = u + v$ ...(i)

$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$ ...(ii)

$\therefore u = x^x$

$$\frac{du}{dx} = x^x\left(1+\ell nx\right) \qquad ...(iii)$$

$\therefore\ v = x^{\sin x}$

$$\frac{dv}{dx} = x^{\sin x}\left[\cos x\ell nx + \frac{\sin x}{x}\right] \qquad ...(iv)$$

From equation (ii), (iii) & (iv)

$$\frac{dy}{dx} = x^x\left(1+\ell nx\right) + x^{\sin x}\left[\cos \ell nx + \frac{\sin x}{x}\right]$$

(vii) $y = x^x + x^{x^x} + x^{x^{x^x}}$

Ans. $x^x\left(1+\ln x\right) + x^{x^x}\left(\frac{x^x}{x} + \left(\ln x\right)x^x\left(1+\ln x\right)\right) + x^{x^{x^x}}\left(\frac{x^{x^x}}{x} + (\ln x)x^{x^x}\left(\frac{x^x}{x} + \left(\ln x\right)x^x\left(1+\ln x\right)\right)\right)$

Sol. Let $y = u + v + w$ ...(i)

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx}$$

$$\frac{du}{dx} = x^x\left(1+\ell nx\right) \rightarrow \{\text{Previous Question}\}$$

$$\frac{dv}{dx} = x^{x^x}\left[x^x\left(1+\ell nx\right)\left(\ell nx\right) + \frac{x^x}{x}\right] + \{\text{Previous Question}\}$$

$\frac{dw}{dx} \Rightarrow$ Now, $\ell nw = x^{x^x}\ell nx$

$$\frac{1}{w}\frac{dw}{dx}\left[x^{x^x}\left(x^x\left(1+\ell nx\right)\ell nx + \frac{x^x}{x}\right)\right] + \frac{x^{x^x}}{x}$$

$$\frac{dw}{dx} = x^{x^{x^x}}\left[x^{x^x}\left\{x^x\left(1+\ell nx\right)\ell nx + \frac{x^x}{x}\right\}\ell nx + \frac{x^{x^x}}{2}\right]$$

From equation (i), all add

Q.7 $f(x) = (\sqrt{x})^{\sqrt{x}}\ e^{x^2}$ find f' (1)

Ans. $\frac{5e}{2}$

Sol. $f(x)=\left(\sqrt{x}\right)^{\sqrt{x}}\, e^{x^2}$ ...(i)

Taking ℓn :-

$\ell n y=\sqrt{x}\,\ell n\sqrt{x}+x^2$ ...(ii)

$\frac{1}{y}\frac{dy}{dx}=\frac{\sqrt{x}}{\sqrt{x}}\times\frac{1}{2\sqrt{x}}+\frac{\ell n\sqrt{x}}{2\sqrt{x}}+2x$ ...(iii)

From equation (i) at $x=1 \Rightarrow y=e$

From equation (iii)

$\frac{dy}{dx}=e\left(\frac{1}{2}+0+2\right)=\frac{5e}{2}$

Q.8 $f(x)=\pi^2+2^x+x^2+x^{1/x}$ find $f'(1)$

Ans. $3+2\ln 2$

Sol. $f(x)=0+2^x\,\ell n2+2x+x^{\frac{1}{x}}\left[\frac{1}{x^2}\ell nx+\frac{1}{x}\frac{1}{x}\right]$

$f'(1)=2\,\ell n2+2+1$

$=3+2\,\ell n2$

Q.9 $f(x)=\prod_{n=1}^{100}(x-n)^{n(101-n)}$ find $\frac{f(101)}{f'(101)}$.

Ans. $\frac{1}{5050}$

Sol $\ell n f(x)=\sum_{n=1}^{100} n(101-n)\ell n(x-n)$ ($\prod$ changes to $\sum$)

diff. w.r.t. x, we get,

$$\frac{f'(x)}{f(x)}=\sum_{n=1}^{100}\frac{n(101-n)}{(x-n)}\ ;\qquad \frac{f'(101)}{f(101)}=\sum_{n=1}^{100}\frac{n(101-n)}{(101-n)}=\sum_{n=1}^{100}n=5050\ ;\ \therefore\frac{f(101)}{f'(101)}=\frac{1}{5050}$$

Q.10 If $y=(\sin x)^{\ln x}\operatorname{cosec}(e^x(a+bx))$ and $a+b=\frac{\pi}{2e}$ then the value of $\frac{dy}{dx}$ at $x=1$ is

Ans $\ln(\sin 1)$

Sol. $y=(\sin x)^{\ln x}\operatorname{cosec}(e^x(a+bx))$; $(a+b)=\frac{\pi}{2e}$ {let $E=(e^x(a+bx))$}

$$\frac{dy}{dx}=(\sin)^{\ln x}(-\operatorname{cosec}E.\cot E)\left\{e^x(b)+(a+bx)e^x\right\}$$

$$+\operatorname{cosec}(e^x(a+bx))(\sin x)^{\ell nx}.\left\{\ln x\cot x+\frac{\ln\sin x}{x}\right\}$$

$$\left.\frac{dy}{dx}\right]_{x=1}=(1)(-1)\cdot(0)\{be+(a+b)e\}+(1)(1)[\ln\sin(1)]=\ln(\sin 1)$$

Q.11 If $y=2^{\log_2 x^{2x}}+\left(\tan\frac{\pi x}{4}\right)^{\frac{4}{\pi x}}$ then $\left.\frac{dy}{dx}\right]_{x=1}$ is :

Ans. 4

Sol. $y=2^{\log_2 x^{2x}}+\left(\tan\frac{\pi x}{4}\right)^{\frac{4}{\pi x}}$

$$=(x^2)^x+\left(\tan\frac{\pi x}{4}\right)^{\frac{4}{\pi x}}$$

$$\frac{dy}{dx}=(x^2)^x\left[\frac{x}{x^2}\cdot 2x+\ln x^2\right]+\left(\tan\frac{\pi x}{4}\right)^{\frac{4}{\pi x}}\left[\frac{4}{\pi x}\cdot\frac{1}{\tan\frac{\pi x}{4}}\sec^2\frac{\pi x}{4}.\frac{\pi}{4}+\ln\left(\tan\frac{\pi x}{4}\right)\left(-\frac{4}{\pi x^2}\right)\right]$$

$$\left.\frac{dy}{dx}\right|_{x=1}=(2+0)+\left[\frac{4}{\pi}\cdot\frac{2\pi}{4}-0\right]=2+2=4$$

Q.12 Let, $y=\left(1+\frac{1}{x}\right)^x+x^{\left(1+\frac{1}{x}\right)}$ then $y'(1)$ equals

Ans. $\ln(4e)$

Sol $y = \left(1+\frac{1}{x}\right)^x + x^{\left(1+\frac{1}{x}\right)}$

$$y' = \left(1+\frac{1}{x}\right)^x \left(\frac{x}{1+\frac{1}{x}} \left\{ -\frac{1}{x^2} \right\} + \log\left(1+\frac{1}{x}\right) \right) + x^{\left(1+\frac{1}{x}\right)} \left[\left(\frac{x+1}{x}\right) \cdot \frac{1}{x} + \ln x \cdot \left(-\frac{1}{x^2}\right) \right]$$

$y'(1) = 2\left[l\text{n}\, 2 + \left(-\frac{1}{2}\right) \right] + 1[2] = 2\, l\text{n}\, 2 - 1 + 2 = 2\, l\text{n}\, 2 + 1 = l\text{n}\,(4e)$ Ans.

2.2 PARAMETRIC DIFFERENTIATION

If $y = f(\theta)$ & $x = g(\theta)$ where θ is a parameter, then

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}.$$

1. If $x = a\cos^3 t$ and $y = a\sin^3 t$, then find the value of $\frac{dy}{dx}$.

Ans. $-\tan t$

Sol. $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3a\sin^2 t\cos t}{-3a\cos^2 t\sin t} = -\tan t$

2. $x = at^2$, $y = 2at$ find $\frac{dy}{dx}$ (t is parameter)

Ans. $1/t$

Sol. $\frac{dy}{dt} = 2a$

$$\frac{dx}{dt} = 2at$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2a}{2at} = \frac{1}{t}$$

3. $x = \cos t + t\sin t$; $y = \sin t - t\cos t$ find $\frac{dy}{dx}$

Ans. $\tan t$

Sol. $\frac{dy}{dt} = \cos t - t(-\sin t) - \cos t = t\sin t$

$$\frac{dx}{dt} = -\sin t + \sin t + t\cos t = t\cos t$$

$$\frac{dy}{dx} = \frac{t\sin t}{t\cos t} = \tan t$$

4. $y = a\tan^3\theta$, $x = a\sec^2\theta$ find $\frac{dy}{dx}$ (θ is parameter)

Ans. $\frac{3}{2}\tan\theta$

Sol. $$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \qquad ...(i)$$

$$\frac{dy}{d\theta} = a\cdot 3\tan^2\theta\cdot\sec^2\theta$$

$$\frac{dx}{d\theta} = a\cdot 2\sec\theta\cdot\sec\theta\cdot\tan\theta = 2a\sec^2\theta\cdot\tan\theta$$

From equation (i)

$$\frac{dy}{dx} = \frac{3a\tan^2\theta\cdot\sec^2\theta}{2a\sec^2\theta\cdot\tan\theta} = \frac{3}{2}\tan\theta$$

5. $x = \sqrt{\cos 2t}\,.\cos t$, $y = \sqrt{\cos 2t}\,.\sin t$ find $\frac{dy}{dx}$ at $t = \frac{\pi}{6}$

Ans. 0

Sol. $$\frac{dy}{dt} = \cos t\sqrt{\cos 2t} + \frac{\sin t\times(-\sin 2t)\times 2}{2\sqrt{\cos 2t}}$$

$$= \frac{\cos t\cdot(\cos 2t) - \sin t\cdot\sin 2t}{\sqrt{\cos 2t}}$$

$$\frac{dy}{dt} = \frac{\cos 3t}{\sqrt{\cos 2t}}$$

$$\frac{dx}{dt} = -\sin t\sqrt{\cos 2t} + \frac{\cos t \times (-\sin 2t) \times 2}{2\sqrt{\cos 2t}}$$

$$\frac{dx}{dt} = \frac{-\sin t \cos 2t - \cos t \sin 2t}{\sqrt{\cos 2t}}$$

$$\frac{dx}{dt} = \frac{-\sin 3t}{\sqrt{\cos 2t}}$$

$$\frac{dy}{dx} = \frac{\cos 3t}{-\sin 3t} = -\cot 3t$$

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{6}} = 0$$

6. $x = \sec\ \theta - \cos\ \theta\ \ y = \sec^n(\theta) - \cos^n(\theta)$ [IIT'89]

Prove that $(x^2+4)\left(\frac{dy}{dx}\right)^2 = n^2(y^2+4)$

Ans.

Sol. $x = \sec\theta - \cos\theta$

$$\frac{dx}{d\theta} = \sec\theta\tan\theta + \sin\theta = \tan\theta(\sec\theta + \cos\theta)$$

$y = (\sec\theta)^n - (\cos\theta)^n$

$$\frac{dy}{d\theta} = n(\sec\theta)^{n-1}\sec\theta\tan\theta + n(\cos\theta)^{n-1}\sin\theta = n\tan\theta[(\sec\theta)^n + (\cos\theta)^n]$$

$$\frac{dy}{dx} = \frac{n\tan\theta[(\sec\theta)^n + (\cos\theta)^n]}{\tan\theta(\sec\theta + \cos\theta)}$$

$$\left(\frac{dy}{dx}\right)^2 = n^2\frac{[(\sec\theta)^n + (\cos\theta)^n]^2}{(\sec\theta + \cos\theta)^2}$$

$$= \frac{n^2\left[\left((\sec\theta)^n - (\cos\theta)^n\right)^2 + 4\right]}{(\sec\theta - \cos\theta)^2 + 4} = \frac{n^2(y^2+4)}{x^2+4} \quad \therefore\ (x^2+4)\left(\frac{dy}{dx}\right)^2 = n^2(y^2+4)$$

2.3 DIFFERENTIATION OF ONE FUNCTION WITH RESPECT TO ANOTHER FUNCTION

If $y = f(x)$ and $z = g(x)$ then derivative of $f(x)$ w.r.t. $g(x)$ is given by

$$\frac{dy}{dz} = \frac{dy}{dx}\cdot\frac{dx}{dz} = \frac{f'(x)}{g'(x)}$$

$\therefore$ Differential coefficient of f(x) w.r.t. g(x) $= \dfrac{derivative\ of\ f(x)\ w.r.t.\ x}{derivative\ of\ g(x)\ w.r.t.\ x} = \dfrac{f'(x)}{g'(x)}$

Find derivative f(x) w.r.t. g(x)

1. $f(x) = \sin x, \quad g(x) = \cos x$

Ans. $-\cot x$

Sol. $\dfrac{df}{dg} = \dfrac{f'(x)}{g'(x)} = \dfrac{\cos x}{-\sin x} = -\cot x$

2. $f(x) = x^2, \quad g(x) = e^x$

Ans. $\dfrac{2x}{e^x}$

Sol. $\dfrac{df}{dg} = \dfrac{f'(x)}{g'(x)} = \dfrac{2x}{e^x}$

3. $f(x) = x^{\sin x}, \quad g(x) = (\sin x)^x$

Ans. $\dfrac{x^{\sin x}\left(\cos x \ln x + \dfrac{\sin x}{x}\right)}{(\sin x)^x\left(\ln(\sin x) + x\cot x\right)}$

Sol. $\dfrac{df}{dg} = \dfrac{x^{\sin x}\left[\cos x\,\ell nx + \dfrac{\sin x}{x}\right]}{(\sin x)^x\left[1\cdot\ell n\sin x + x\cot x\right]}$

4. $f(x) = \sin^{-1}(x), \ g(x) = \cos^{-1}(x)$

Ans. -1

Sol. $\dfrac{df}{dg} = \dfrac{f'(x)}{g'(x)} = \dfrac{\dfrac{1}{\sqrt{1-x^2}}}{\dfrac{-1}{\sqrt{1-x^2}}} = -1$

5. $f(x) = \cos^{-1}(2x^2-1), \quad g(x) = \cos^{-1}x \ \forall \ \ x > 0$

Ans. 2

Sol. $f(x) = 2\cos^{-1}x$ [Property II]

$$\frac{df}{dg} = \frac{\frac{-1}{\sqrt{1-x^2}}}{\frac{-1}{\sqrt{1-x^2}}} = 2$$

6. $f(x) = \cos^{-1}(2x^2-1)$, $g(x) = \sqrt{1-x^2}$ at $x = \frac{1}{2}$

Ans. 4

Sol. $f(x) = 2\cos^{-1}x$, $g(x) = \sqrt{1-x^2}$

$$\frac{df}{dg} = \frac{-\frac{2}{\sqrt{1-x^2}}}{\frac{(-2x)}{2\sqrt{1-x^2}}} = \frac{2}{x}\Bigg|_{x=\frac{1}{2}} = \frac{2}{\frac{1}{2}} = 4$$

7. $f(x) = e^{\sin^{-1}(x)}$, $g(x) = e^{-\cos^{-1}(x)}$

Ans. $e^{\pi/2}$

Sol. $$\frac{df}{dg} = \frac{e^{\sin^{-1}x} \times \frac{1}{\sqrt{1-x^2}}}{e^{-\cos^{-1}x} \times \frac{1}{\sqrt{1-x^2}}} = e^{\sin^{-1}x+\cos^{-1}x}$$

$$= e^{\frac{\pi}{2}}$$

8. $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ at $x = \frac{-1}{3}$

Ans. -1

Sol. $f(x) = 2\tan^{-1}x$; $g(x) = -2\tan^{-1}x$

$$\frac{df}{dg}=\frac{f'(x)}{g'(x)}=\frac{\frac{2}{1+x^2}}{-\frac{2}{1+x^2}}=-1$$

2.4 DERIVATIVE OF IMPLICIT FUNCTIONS

For example $x^3+y^3+3axy+c = 0$, $x^y+y^x = a^b$ etc.

Working rule for finding the derivative

(i) Differentiate every term of f(x,y) = 0 with respect to x.

(ii) Collect the coefficients of dy/dx and obtain the value of dy/dx.

1. $e^{x-y} = x^y$; prove that $\frac{dy}{dx}=\frac{\ln x}{(1+\ln x)^2}$

Sol. $e^{x-y} = x^y$ ...(i)

$x - y = y\ell nx$ [In differentiation log base e always]

$x = y + y\,\ell nx$

$y(1 + \ell nx) = x$

$$y=\frac{x}{(1+\ell nx)}$$

On differentiation w.r.t. x

$$\frac{dy}{dx}=\frac{(1+\ell nx)-x\times\frac{1}{x}}{(1+\ell nx)^2}$$

$$\frac{dy}{dx}=\frac{1+\ell nx-1}{(1+\ell nx)^2}$$

$$\frac{dy}{dx}=\frac{\ln x}{(1+\ln x)^2}$$

Hence proved.

2. $\sin y = x \sin (a + y)$; prove that $\frac{dy}{dx}=\frac{\sin^2(a+y)}{\sin a}$

Ans.

Sol. $\sin y = x \sin (a+y)$

$$\therefore\ x = \frac{\sin y}{\sin (a+y)}$$

differentiating the function with respect to y

$$\frac{dx}{dy} = \frac{\sin(a+y)\cos y - \sin y \cos(a+y)}{\sin^2(a+y)}$$

$$= \frac{\sin(a+y-y)}{\sin^2(a+y)} = \frac{\sin a}{\sin^2(a+y)}$$

$$\therefore \frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$$

3. $x\sqrt{1+y} + y\sqrt{1+x} = 0$; prove that $\frac{dy}{dx} = \frac{-1}{(1+x)^2}$

Ans.

Sol. $x\sqrt{1+y} + y\sqrt{1+x} = 0$

$$x\sqrt{1+y} = -y\sqrt{1+x}$$

$$x^2(1+y) = y^2(1+x)$$

$$x^2 + x^2y = y^2 + y^2x$$

$$x^2 - y^2 = y^2x - x^2y$$

$$(x-y)(x+y) = xy(y-x)$$

$$x + y = -xy$$

$$y(1+x) = -x$$

$$y = \frac{-x}{1+x}$$

$$\frac{dy}{dx} = \frac{(1+x)(-1)+x(1)}{(1+x)^2}$$

$$\frac{dy}{dx} = \frac{-1}{(1+x)^2}$$

4. $ax^2 + 2h\,xy + by^2 = 0$; prove that $\frac{dy}{dx} = \frac{y}{x}$

Ans.

Sol $2ax + 2by\,y' + 2h(xy' + y) = 0$

$$(ax + hy) + (by + hx)y' = 0$$

$$y' = -\frac{ax+hy}{by+hx}$$

$$ax^2 + hxy + hxy + by^2 = 0$$

$$x(ax + hy) + y(by + hx) = 0; \qquad \therefore \quad -\frac{ax+hy}{by+hx} = \frac{y}{x}$$

Note: For any homogeneous function of degree n, $\frac{dy}{dx} = \frac{y}{x}$ always, and $\frac{d^2y}{dx^2} = 0$

5. $\ln(x+y) = x\,e^y$; find $\left(\frac{dy}{dx}\right)_{(0,1)}$

Ans $e-1$

Sol Differentiating

$$\frac{1}{(x+y)}\left(1+\frac{dy}{dx}\right) = x\,e^y\frac{dy}{dx} + e^y \;;\; \text{put } x=0 \text{ and } y=1, \text{ we get}$$

$$1+\frac{dy}{dx}\Big]_{(0,1)} = e \quad \Rightarrow \quad \left(\frac{dy}{dx}\right)_{(0,1)} = e-1$$

6. $y^5 + xy^2 + x^3 = 4x + 3$; find $\left(\frac{dy}{dx}\right)_{(2,1)}$

Ans. -1

Sol. $y^5 + xy^2 + x^3 = 4x + 3$ ...(i)

On differentiating w.r.t. x :-

$$5y^4\frac{dy}{dx} + x\cdot 2y\frac{dy}{dx} + y^2 + 3x^2 = 4$$

Put $x = 2, y = 1$

$$\frac{5dy}{dx} + \frac{4dy}{dx} = 4 - 13 \Rightarrow \frac{dy}{dx} = -1$$

7. $y = \sqrt{x+\sqrt{x+\sqrt{x+.....\infty}}}$

(i) Prove that $\frac{dy}{dx} = \frac{1}{2y-1}$ (ii) Prove that $\frac{dy}{dx} = \frac{y}{y^2+x}$

Sol. $y = \sqrt{x+\sqrt{x+\sqrt{x+.....\infty}}}$

$y^2 = x + y$ _____(i)

$\frac{dy}{dx} = \frac{1}{2y-1}$ _____(ii)

Multiply & divide by y

$$\frac{dy}{dx} = \frac{y}{2y^2 - y}$$

$$\frac{dy}{dx}=\frac{y}{y^2+\left(y^2-y\right)}$$

From equation (i)

$$\frac{dy}{dx}=\frac{y}{y^2+x}$$

8. $y=\cfrac{x}{1+\cfrac{x}{1+\cfrac{x}{1+\cfrac{x}{1+.....\infty}}}}$ prove that $\frac{dy}{dx}=\frac{1}{2y+1}$

Sol. $y=\frac{x}{1+y}$

$y+y^2=x$

$$\frac{dy}{dx}+2y\frac{dy}{dx}=1$$

$$\frac{dy}{dx}=\frac{1}{2y+1}$$

9. $y=x^{x^{x^{\cdots\cdots\infty}}}$ prove that $\frac{dy}{dx}=\frac{y^2}{x(1-y\ln x)}$

Sol. $y=x^{x^{x^{\cdots\cdots\infty}}}$

$y=x^y$

$\ell n\, y=y\,\ell n\, x$

$$\frac{1}{y}\frac{dy}{dx}=\ell nx\frac{dy}{dx}+\frac{y}{x}$$

$$\frac{dy}{dx}=\frac{y^2}{x(1-y\ln x)}$$

10. $y=(\ln x)^{\ln x^{\ln x^{\cdots\cdots\infty}}}$ prove that $\frac{dy}{dx}=\frac{y^2}{x\ln x(1-y\ln(\ln x))}$

Sol. $y=(\ell n\, x)^y$

$\ell n\, y = y\, \ell n\,(\ell n\, x)$

$$\frac{1}{y}\frac{dy}{dx} = \ell n\left(\ell n\, x\right)\frac{dy}{dx} + \frac{y}{x\ell n\, x}$$

$$\frac{dy}{dx} = \frac{y^2}{x\ell nx\left[1 - y\ell n\left(\ell nx\right)\right]}$$

11. $\sqrt{1-x^2} + \sqrt{1-y^2} = a\,(x-y)$; prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

Sol $x = \sin\theta$

$y = \sin\phi$

$x, y \in [-1, 1]$ and θ & $\phi \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\cos\theta + \cos\phi = a\,(\sin\theta - \sin\phi)$

$$\tan\left(\frac{\theta-\phi}{2}\right) = \frac{1}{a}$$

$$\frac{\theta-\phi}{2} = \tan^{-1}\left(\frac{1}{a}\right)$$

$$\sin^{-1}x - \sin^{-1}y = 2\tan^{-1}\left(\frac{1}{a}\right)$$

$$\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

12. If $\sqrt{y-\sqrt{y-\sqrt{y-\sqrt{y.....\infty}}}} = \sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x.....\infty}}}}$. Then prove that $\frac{dy}{dx} = \frac{x-y-1}{x-y+1}$.

Sol. $\left(y+x-1\right) = \sqrt{1+4x}$

$y^2 + x^2 + 1^2 - 2xy + 2x - 2y = 1 + 4x$

$x^2 + y^2 - 2xy - 2x - 2y = 0$

$$\frac{dy}{dx} = -1\left\{\frac{2x-2y-2}{2y-2x-2}\right\}$$

$$\frac{dy}{dx} = \frac{-2}{-2}\left\{\frac{x-y-1}{x-y+1}\right\}$$

$$\frac{dy}{dx}=\frac{x-y-1}{x-y+1}$$

2.5 DERIVATIVE OF INVERSE OF A FUNCTION

If g is inverse of f, then $g\{f(x)\}=x$

Differentiate wrt x

$g'\{f(x)\}f'(x)=1$

Q.1. $f(x)=x^5+x^3$ $g(x)=f^{-1}(x)$ find $g'(2)$

Ans 1/8

Sol. $(gof)(x)=x$

$g'[f(x)]\,f'(x)=1$;

when $f(x)=2$ then $x=1$

$g'(2)\cdot f'(1)=1$ (but $f'(1)=8$)

$\therefore$ $g'(2)=1/8$

Q.2. $f(x)=e^{x+x^2+x^3}$ $g(x)=f^{-1}(x)$ find $g'(e^3)$

Ans. $\frac{1}{6e^3}$

Sol. $f(x)=e^{x+x^2+x^3}$ $f'(x)=e^{x+x^2+x^3}\left\{1+2x+3x^2\right\}$

$g(x)=f'(x)$ $f(1)=e^3.6$

$g'(e^3)=?$

$e^{x+x^2+x^3}=e^3$

$\Rightarrow x=1$

$$g'\left(e^3\right)=\frac{1}{f'(1)}$$

$$g'\left(e^3\right)=\frac{1}{6e^3}$$

Q.3 $f(x)=\frac{1}{1+x^n}$, $g(x)=f^{-1}(x)$ prove that $g'(x)=1+(g(x))^n$

Sol. Since g is the inverse of f. Therefore

$f(g(x))=x$ for all x

$\Rightarrow \frac{d}{dx}f(g(x))=1$ for all x

$\Rightarrow f'(g(x))\, g'(x) = 1 \quad \Rightarrow \quad g'(x) = \frac{1}{f'(g(x))} = 1 + (g(x)^n$

3. HIGHER ORDER DERIVATIVES

Let a function $y = f(x)$ be defined on an open interval (a, b). It's derivative, if it exists in (a, b) is a certain function $f'(x)$ [or (dy/dx) or y'] & is called the first derivative of y w. r. t. x.

If it happens that the first derivative has a derivative in (a, b) then this derivative is called the second derivative of y w. r. t. x & is denoted by $f''(x)$ or (d^2y/dx^2) or y''. While the first derivative denotes slope of the graph, the second derivative denotes it's concavity.

Similarly, the 3rd order derivative of y w. r. t. x, if it exists, is defined by $\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right)$, it is also denoted by $f'''(x)$ or y'''.

1. $y = \sin x$ prove that $y_2 + y = 0$

Sol. $y = \sin x$ ...(i)

$y_1 = \cos x$ ...(ii) {differentiating w.r.t. x}

$y_2 = -\sin x$ ...(iii) {differentiating w.r.t. x}

From equation (i) + (iii)

$y_2 + y = 0$

2. $y = e^{2x}$ find (i) $\frac{y_3}{y}$ (ii) Prove that $y_2 = 4y$

Ans. (i) 8

Sol. (i) $y = e^{2x}$ _____(i)

$y_1 = 2e^{2x}$ _____(ii)

$y_2 = 4e^{2x}$ _____(iii)

$y_3 = 8e^{2x}$ _____(iv)

(ii) From equation (iii)

$y_2 = 4y$

$\frac{y_3}{y} = \frac{8 \cdot e^{2x}}{e^{2x}} = 8$

3. $y = \sin x$ find $\frac{d^2y}{dx^2}$ & $\frac{d^2x}{dy^2}$ (in terms of x)

Ans. $\frac{d^2y}{dx^2} = -\sin x, \quad \frac{d^2x}{dy^2} = \sec^2 x \tan x$

Sol. (i) $y = \sin x$ _____(i)

$\frac{dy}{dx} = \cos x$ _____(ii)

$$\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{dy}{dx}\cos x$$

$$\frac{d^2y}{dx^2}=-\sin x \qquad \text{_____(iii)}$$

(ii) From equation (ii)

$$\frac{dx}{dy}=\sec x$$

$$\frac{d}{dy}\left(\frac{dx}{dy}\right)=\frac{d}{dy}\sec x$$

$$\frac{d^2x}{dy^2}=\sec x\cdot\tan\frac{dx}{dy}$$

$$=(\sec x\cdot\tan x)\sec x$$

$$\frac{d^2x}{dy^2}=\sec^2 x\tan x$$

4. $y = \sin(\sin x)$ prove that $y_2 + (\tan x)\, y_1 + y\cos^2 x = 0$

Sol Such expression can be easily proved using implicit differentiation

$\Rightarrow$ $y' = \cos(\sin x)\cos x$

$\Rightarrow$ $\sec x.y' = \cos(\sin x)$

again differentiating w.r.t x, we get

$\sec x\, y'' + y' \sec x \tan x = -\sin(\sin x)\cos x$

$\Rightarrow$ $y'' + y' \tan x = -y\cos^2 x$

$\Rightarrow$ $y'' + (\tan x)\, y' + y\cos^2 x = 0$

5. $x = \cos t + t\sin t$ $\quad y = \sin t - t\cos t$; find $\frac{d^2y}{dx^2}$ & $\frac{d^2x}{dy^2}$ (in terms of t)

Ans $\frac{d^2x}{dy^2}=-\frac{\operatorname{cosec}^3 t}{t}$; $\frac{d^2y}{dx^2}=\frac{\sec^3 t}{t}$

Sol. $x = \cos t + t\sin t$ ...(i)

$$\frac{dx}{dt} = -\sin t + t\cos t + \sin t$$

$$\frac{dx}{dt} = t\cos t \qquad ...(ii)$$

Also, $y = \sin t - t\cos t$ ...(iii)

$$\frac{dy}{dt} = \cos t - t(-\sin t) - \cos t$$

$$\frac{dy}{dt} = t\sin t \qquad ...(iv)$$

$\therefore$ From equation (iv) & (ii)

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{t\sin t}{t\cos t} = \tan t$$

$$\frac{dy}{dx} = \tan t \qquad ...(v)$$

(i) $$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(\tan t) = \sec^2 t \cdot \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \sec^2 t \times \frac{1}{t\cos t}$$

$$\frac{d^2y}{dx^2} = \frac{\sec^3 t}{t}$$

(ii) $$\frac{d}{dy}\left(\frac{dx}{dy}\right) = \frac{d}{dy}(\cot t) = -\text{cosec}^2 t\frac{dt}{dy}$$

$$= -\text{cosec}^2 t \cdot \frac{1}{t\sin t}$$

$$= -\frac{\text{cosec}^3 t}{t}$$

6. $x^3 + 3x^2y - 6xy^2 + 2y^3 = 0$; prove that $\frac{d^2y}{dx^2} = 0$

Sol. for any homogeneous function of degree n

$$\frac{d^2y}{dx^2} = 0$$

7. $e^{x+y} = y^2$ prove that $y_2 = \frac{2y}{(2-y)^3}$

Sol. $e^{x+y} = y^2$ ...(i)

$\ell n e^{x+y} = \ell n y^2$ $\quad \therefore y_2 = \frac{2y}{(2-y)^3}$

$x + y = 2\ell ny$

$$1 + y_1 = \frac{2}{y} y_1$$

$$1 = \left(\frac{2}{y} - 1\right) y_1$$

$$y_1 = \frac{y}{(2-y)} \qquad \text{...(ii)}$$

$\therefore$ Again differentiating equation (ii) w.r.t. x :

$$y_2 = \frac{(2-y)y_1 - y(-y_1)}{(2-y)^2}$$

$$y_2 = \frac{y_1(2-y+y)}{(2-y)^2} = \frac{2y_1}{(2-y)^2}$$

$$y_2 = \frac{2 \cdot \frac{y}{(2-y)}}{(2-y)^2} = \frac{2y}{(2-y)^3}$$

8. Prove that $\frac{d^2x}{dy^2} = \frac{\frac{-d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3}$

Sol. $\frac{d^2x}{dy^2} = \frac{d}{dy}\left(\frac{dx}{dy}\right)$

$= \frac{d}{dy}\left(\frac{1}{\frac{dy}{dx}}\right)$

$= \frac{-1}{\left(\frac{dy}{dx}\right)^2}\left(\frac{d}{dy}.\frac{dy}{dx}\right)$

$\frac{d^2x}{dy^2} = -\left[\frac{\frac{dx}{dy}\left(\frac{d}{dx}.\frac{dy}{dx}\right)}{\left(\frac{dy}{dx}\right)^2}\right]$

$= \frac{-\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^2 \times \left(\frac{dy}{dx}\right)} = \frac{-\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx}\right)^3}$

9. $y = e^{\cos^{-1}(x)}$ prove that $(1 - x^2)y_2 - x\, y_1 - y = 0$

Sol. $y_1 = e^{\cos^{-1} x} \times \frac{-1}{\sqrt{1-x^2}} = \frac{-y}{\sqrt{1-x^2}}$ _____(i)

Again differentiating w.r.t. x

$$y_2 = \frac{\sqrt{1-x^2}\cdot(-y_1) + y\left(\frac{-2x}{2\sqrt{1-x^2}}\right)}{\left(1-x^2\right)}$$

$$(1-x^2)y_2 = -y_1\sqrt{1-x^2} - \frac{xy}{\sqrt{1-x^2}}$$

$$(1-x^2)y_2 = y + xy_1$$

$$(1-x^2)y_2 - x\,y_1 - y = 0$$

10. $y = \left(\frac{1}{x}\right)^x$ prove that $y_2(1) = 0$

Sol Now taking $\log_e$ both sides, we get

$\ell n\, y = -x\, \ell n\, x$ when $x = 1$, then $y = 1$

$\ell n\, y = -x\, \ell n\, x$

$\Rightarrow \quad \frac{y'}{y} = -(1 + \ell n\, x) \quad \Rightarrow \quad y' = -y(1 + \ell n\, x) \quad$ .....(i)

again differentiating. w.r.t. to x,

$y'' = -y'(1 + \ell n\, x) - y \cdot \frac{1}{x} \quad \Rightarrow \quad y'' = y(1 + \ell n\, x)^2 - \frac{y}{x}$ (using (i))

$\Rightarrow \quad y''(1) = 0$

11. $y^2 = 4ax$. prove that $\frac{d^2y}{dx^2} \times \frac{d^2x}{dy^2} = \frac{-2a}{y^3}$

Sol. On differentiating w.r.t. x

$$2y\cdot\frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{2a}{y} \qquad \text{...(ii)}$$

For $\frac{d^2y}{dx^2}$

$$\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{d}{dx}\left(\frac{2a}{y}\right)=\frac{-2a}{y^2}\frac{dy}{dx}$$

$$\frac{d^2y}{dx^2}=\frac{-2a}{y^2}\times\frac{2a}{y}=\frac{-4a^2}{y^3} \qquad ...(iii)$$

$\therefore$ For $\frac{d^2x}{dy^2}$

$$\frac{d}{dy}\left(\frac{dx}{dy}\right)=\frac{d}{dy}\left(\frac{y}{2a}\right)=\frac{1}{2a} \qquad ...(iv)$$

From equation (iii) & (iv)

$$\frac{d^2y}{dx^2}\times\frac{d^2x}{dy^2}=\frac{-4a^2}{y^3}\times\frac{1}{2a}=-\frac{2a}{y^3}$$

12. Use the substitution $x = \tan\theta$ to show that the equation.

$$\frac{d^2y}{dx^2}+\frac{2x}{1+x^2}\frac{dy}{dx}+\frac{y}{(1+x^2)^2}=0 \text{ changes to } \frac{d^2y}{d\theta^2}+y=0.$$

Sol.

13. If $y^{\frac{1}{m}} + y^{-\frac{1}{m}} = 2x$ then prove that $(x^2 - 1)y_3 + 3xy_2 + (1 - m^2)y_1 = 0$

Sol. $$\frac{1}{m}y^{\left(\frac{1}{m}-1\right)}\cdot y_1-\frac{1}{m}y^{\left(-\frac{1}{m}-1\right)}\cdot y_1=2$$

$$y^{\frac{1}{m}}-y^{-\frac{1}{m}}=\frac{2my}{y_1}$$

$$\left(y^{\frac{1}{m}}-y^{-\frac{1}{m}}\right)^2=\left(\frac{2my}{y_1}\right)^2$$

$$4x^2-4=\frac{4m^2y^2}{(y_1)^2}$$

$$4(x^2-1)y_1^2=4m^2y^2$$

$(x^2 - 1)2y_1y_2 + y_1^2 \cdot 2x = m^2\, 2y\, y_1$

$2(x^2 - 1)y_2 + 2xy_1 = 2m^2y$

$2(x^2 - 1)y_3 + 4xy_2 + 2xy_2 + 2y_1 = 2m^2y_1$

$(x^2 - 1)y_3 + 3xy_2 + (1 - m^2)y_1 = 0$

14. If $y = \tan^{-1}\left(\sec x^3 - \tan x^3\right), \frac{\pi}{2} < x^3 < \frac{3\pi}{2}$, then prove that $x^2y'' - 6y + \frac{3\pi}{2} = 0$

Sol. $y = \tan^{-1}\left(\frac{1}{\cos x^3} - \frac{\sin x^3}{\cos x^3}\right)$

$y = \tan^{-1}\left(\frac{1-\sin x^3}{\cos x^3}\right)$

$y = \tan^{-1}\left(\frac{1-\cos\left(\frac{\pi}{2} - x^3\right)}{\sin\left(\frac{\pi}{2} - x^3\right)}\right)$

$y = \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{x^3}{2}\right)\right)$

$-\frac{\pi}{2} < \frac{\pi}{4} - \frac{x^3}{2} < 0$

$y = \frac{\pi}{4} - \frac{x^3}{2}$

$\frac{dy}{dx} = \frac{-3x^2}{2}$

$y' = \frac{-3}{2}x^2$

$y'' = -3x$

Now, $x^2y'' - 6y + \frac{3\pi}{2}$

$x^2(-3x) - 6\left(\frac{\pi}{4} - \frac{x^3}{2}\right) + \frac{3\pi}{2} = 0$

Hence proved

15. Let f(x) be polynomial function such that $f(x) + f'(x) + f''(x) = x^5 + 64$. Then, the value of $\lim_{x\to 1}\frac{f(x)}{x-1}$ is equal to:

Ans. –15

Sol. f(x) = polynomial

$f(x)+f'(x)+f''(x)=x^5+64$

$\lim_{x\to 1}\frac{f(x)}{x-1}$ f(1) = 0 होना चाहिए।

$\lim_{x\to 1}\frac{f(x)}{1}$ f'(1) होना चाहिए।

$f(x)+f'(x)+f''(x)=x^5+64$

$f'(x)+f''(x)+f'''(x)=5x^4$

$f'''(x)+f''x+f'x=60x^2$

$f''(x)-f''(x)=20x^3-60x^2$

$f''(1)-f'(1)=20-60=-40$

$f''(1)-120=-40$

$f''(1)=80$

$f(x)+f'(x)+f''(x)=x^5+64$

$f(1)+f'(1)+f''(1)=65$

$f'(1)=-15$

4 ANALYSIS OF ITF

Q.1 $y=f(x)=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

(i) Find Domain and Range.

Sol $y=f(x)=\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\begin{cases}\pi-2\tan^{-1}x; & x>1\\ 2\tan^{-1}x; & -1\le x\le 1\\ -\pi-2\tan^{-1}x; & x<-1\end{cases}$

Domain is $x\in R$ & **Note:** f is odd, aperiodic bound

range is $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

(ii) Find value/limit of y at boundary points of domain and point where definition is changing.

Sol. $\lim_{x\to-\infty}f(x)=-\pi-2\tan^{-1}(-\infty)$

$=-\pi-2\left(\frac{\pi}{2}\right)$

$= -\pi + \pi = 0$

At $x = -1 = 2\tan^{-1}(-1)$

$$2\times -\frac{\pi}{4} = -\frac{\pi}{2}$$

At $x = 1 = 2\tan^{-1}(1) = 2\times\frac{\pi}{4} = \frac{\pi}{2}$

At $x \to \infty$, $\lim\limits_{x\to\infty} f(x) = \pi - 2\tan^{-1}(\infty)$

$$= \pi - 2\cdot\frac{\pi}{2} = 0$$

(iii) Check continuity.

Sol. Continuous for $\forall\ x \in R$

(iv) Check differentiability.

Sol $$\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for} \quad |x|<1 \\ \text{nonexistent} & \text{for} \quad |x|=1 \\ -\frac{2}{1+x^2} & \text{for} \quad |x|>1 \end{cases}$$

f is not diff. at $x = 1, -1$

(v) Find intervals of monotonocity.

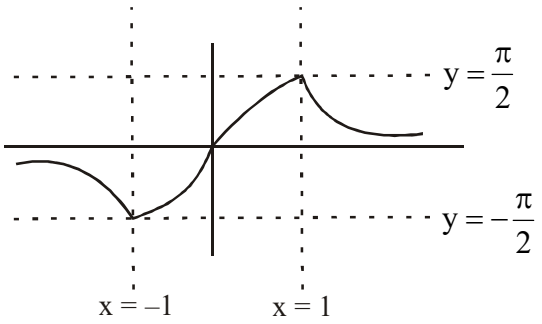
Sol. I in $(-1, 1)$ & D in $(-\infty, -1) \cup (1, \infty)$

(vi) Find $f''(x)$ and determine nature of concavity.

Sol. $$f''(x) = \begin{cases} \frac{+4x}{(1+x^2)^2} & ; \quad x > 1 \\ \frac{-4x}{(1+x^2)^2} & ; \quad -1 < x < 1 \\ \frac{4x}{(1+x^2)^2} & ; \quad x < -1 \end{cases}$$

(vii) Plot graph of f(x)

Sol.



Q.2 $y = f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

Sol. $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x; & x \geq 0 \\ -2\tan^{-1}x; & x < 0 \end{cases}$

(i) Domain & range :-

$Df \in R$ & Range $\in [0, \pi]$

(ii) Values/limit

$$\lim_{x\to-\infty} f(x) = \pi \Rightarrow \lim_{x\to\infty} f(x) = \pi$$

$$\lim_{x\to 0} f(x) = 0$$

(iii) Continuity :-

LHL = f(0) = RHL = 0

$\therefore$ Continue at x = 0

(iv) Differentiability :-

$$f'(x) = \begin{cases} \frac{2}{1+x^2}; & x > 0 \\ \frac{-2}{1+x^2}; & x < 0 \end{cases}$$

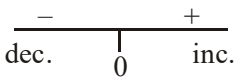
Not differentiable at x = 0

sharp corner at x = 0

(v) Interval of monotonocity

IOI $x \in (0, \infty)$

IOD $x \in (-\infty, 0)$



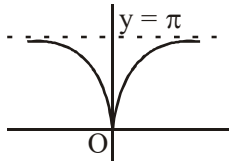
(vi) Concavity

$$f''(x) \begin{cases} \frac{-4x}{\left(1+x^2\right)^2} & ; \quad x \geq 0 \\ \frac{4x}{\left(1+x^2\right)^2} & ; \quad x < 0 \end{cases}$$

$x \in (-\infty, 0) \rightarrow f''(x) < 0$, concave down

$x \in (0, \infty) \rightarrow f''(x) < 0$, concave down

(vii) Graph



Q.3 $y = f(x) = \tan^{-1}\dfrac{2x}{1-x^2}$

Sol. $$f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \begin{cases} \pi + 2\tan^{-1}x & ; \quad x < -1 \\ 2\tan^{-1}x & ; \quad -1 < x < 1 \\ 2\tan^{-1}x - \pi & ; \quad x > 1 \end{cases}$$

(i) Domain & Range :-

$Df \in R - \{-1, 1\}$, Range $\in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

(ii) Values/limit

$\lim\limits_{x \to -\infty} f(x) = 0$

$\therefore$ At $x = -1$

L.H.L. R.H.L.

$\lim\limits_{x \to (-1)^-} f(x) = \dfrac{\pi}{2}$ $\lim\limits_{x \to (-1)^+} f(x) = -\dfrac{\pi}{2}$

At $x \to \infty$,

$\lim\limits_{x \to \infty} f(x) = 0$

(iii) Continuity :-

Discontinue at $x = -1, 1$

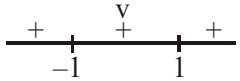
(iv) Differentiability

$$f'(x) = \begin{cases} \dfrac{2}{1+x^2} & ; \quad x < -1 \\ \dfrac{2}{1+x^2} & ; \quad -1 < x < 1 \\ \dfrac{2}{x+x^2} & ; \quad x > 1 \end{cases}$$

$\therefore$ Not differentiable at $x = 1, -1$

As discontinue at $x = -1, 1$

(v) Interval of monotonocity



I.O.I. $x \in (-\infty, -1) \cup (-1, 1) \cup (1, \infty) \Rightarrow x \in R - \{-1, 1\}$

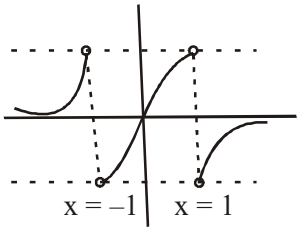
(vi) Concavity

$$f''(x)=\frac{-4x}{\left(1+x^2\right)^2};\ x\in R-\{-1,1\}$$

For $x\in(-\infty,0)$, $f''(x)>0$ concave up

For $x\in(0,\infty)$, $f''(x)<0$ concave down

(vii) Graph



Q.4 $y = f(x) = \sin^{-1}(3x - 4x^3)$

Sol.

$$f(x)=\sin^{-1}\left(3x-4x^3\right)=\begin{cases}-\pi-3\sin^{-1}x & ;\ -1\le x<-\frac{1}{2}\\ 3\sin^{-1}x & ;\ -\frac{1}{2}\le x\le\frac{1}{2}\\ \pi-3\sin^{-1}x & ;\ \frac{1}{2}<x\le 1\end{cases}$$

(i) Domain & Range :-

$D\ f\in[-1,1]$; Range $\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

(ii) Values/limit :-

$f(-1) = -\pi - 3\sin^{-1}(-1)$

$$=-\pi-3\left(-\frac{\pi}{2}\right)=\frac{\pi}{2}$$

$$f\left(-\frac{1}{2}\right)=-\frac{\pi}{2}$$

$$f\left(\frac{1}{2}\right)=\frac{\pi}{2}$$

$$f(1)=-\frac{\pi}{2}$$

(iii) Continuity :-

Continue for $\forall\ x\in[-1,1]$

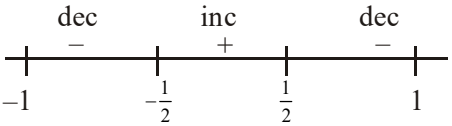
(iv) Differentiability :-

$$f'(x)=\begin{cases}\dfrac{-3}{\sqrt{1-x^2}\ +\ +ve\sqrt{\ }\ \text{Quantity}} & ;\quad -1\le x<-\dfrac{1}{2}\\ \dfrac{3}{\sqrt{1-x^2}} & ;\quad -\dfrac{1}{2}\le x\le\dfrac{1}{2}\\ \dfrac{-3}{\sqrt{1-x^2}} & ;\quad \dfrac{1}{2}<x<1\end{cases}$$

$\therefore$ Not differentiable at $x=-\dfrac{1}{2},\dfrac{1}{2}$

Sharp corner at $x=-\dfrac{1}{2},\dfrac{1}{2}$

(v) Interval of monotonocity :-



IOD $x\in\left[-1,-\dfrac{1}{2}\right)\cup\left(\dfrac{1}{2},1\right]$

IOI $x\in\left(-\dfrac{1}{2},\dfrac{1}{2}\right)$

(vi) Concavity

$$f''(x)=\begin{cases}-\dfrac{3x}{\left(\sqrt{1-x^2}\right)^{\frac{3}{2}}} & ;\quad x\in\left[-1,-\dfrac{1}{2}\right)\cup\left(\dfrac{1}{2},1\right]\\ \dfrac{3x}{\left(\sqrt{1-x^2}\right)^{\frac{3}{2}}} & ;\quad x\in\left(-\dfrac{1}{2},\dfrac{1}{2}\right)\end{cases}$$

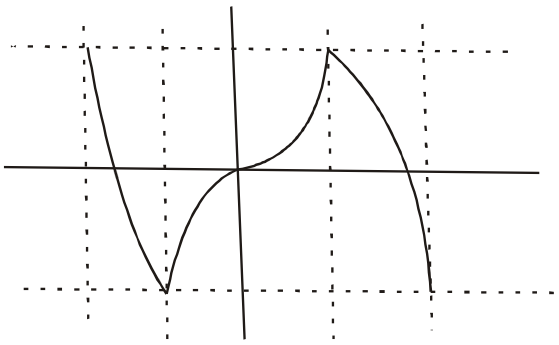
For $x\in\left[-1,\dfrac{1}{2}\right)$ $f''(x)>0$, concave up

$x\in\left(\dfrac{1}{2},1\right]$, $f''(x)<0$, concave down

$x\in\left(-\dfrac{1}{2},0\right)$, $f''(x)<0$, concave down

$x\in\left(0,\dfrac{1}{2}\right)$, $f''(x)>0$, concave up

(vii) Graph :-



Q.5 $y = f(x) = \cos^{-1}(4x^3 - 3x)$

Sol.

$$f(x) = \cos^{-1}(4x^3 - 3x) = \begin{cases} 3\cos^{-1} x - 2\pi & ; \quad -1 \le x < -\frac{1}{2} \\ 2\pi - 3\cos^{-1} x & ; \quad -\frac{1}{2} \le x \le \frac{1}{2} \\ 3\cos^{-1} x & ; \quad \frac{1}{2} < x \le 1 \end{cases}$$

(i) Domain & Range :-

D f $\in [-1,1]$ & Rf $\in [0, \pi]$

(ii) Values/limits :-

$f(-1) = \pi$, $f\left(-\frac{1}{2}\right) = 0$

$f\left(\frac{1}{2}\right) = \pi$, $f(1) = 0$

(iii) Continuity :-

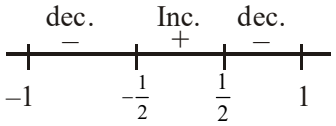
Continue $\forall x \in [-1,1]$

(iv) Differentiability

$$f'(x) = \begin{cases} \frac{-3}{\sqrt{1-x^2}} & ; \quad -1 \le x < -\frac{1}{2} \\ \frac{3}{\sqrt{1-x^2}} & ; \quad -\frac{1}{2} \le x \le \frac{1}{2} \\ \frac{-3}{\sqrt{1-x^2}} & ; \quad \frac{1}{2} < x \le 1 \end{cases}$$

$\therefore$ Not differentiable at $x = -\frac{1}{2}, \frac{1}{2}$

(v) Monotonocity



I.O.I. $x \in \left(-\frac{1}{2}, \frac{1}{2}\right)$

I.O.D. $x \in \left[-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right]$

(vi) Concavity

$$f''(x) = \begin{cases} \dfrac{-3x}{\left(1-x^2\right)^{\frac{3}{2}}}; & x \in \left[-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right] \\ \dfrac{3x}{\left(1-x^2\right)^{\frac{3}{2}}}; & x \in \left(-\frac{1}{2}, \frac{1}{2}\right) \end{cases}$$

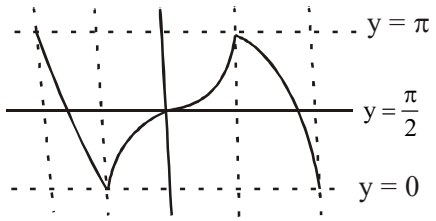
$\therefore$ For $x \in \left[-1, -\frac{1}{2}\right)$, $f''(x) > 0$ concave up

$x \in \left(-\frac{1}{2}, 0\right)$, $f''(x) < 0$ concave down

$x \in \left(0, \frac{1}{2}\right)$, $f''(x) > 0$ concave up

$x \in \left(\frac{1}{2}, 1\right]$, $f''(x) < 0$, concave down

(vii) Graph :-



Q.6 $y = \tan^{-1}\dfrac{3x - x^3}{1 - 3x^2}$

Sol. $$f(x)=\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)=\begin{cases}3\tan^{-1}x & ; \quad -\frac{1}{\sqrt{3}}<x<\frac{1}{\sqrt{3}}\\ -\pi+3\tan^{-1}x & ; \quad x>\frac{1}{\sqrt{3}}\\ \pi+3\tan^{-1}x & ; \quad x<-\frac{1}{\sqrt{2}}\end{cases}$$

(i) Domain & Range :-

$$Df\in R\left\{-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right\} \ \& \ Rf\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$$

Domain

(ii) Values/limits :-

$$\lim_{x\to-\infty}f(x)=-\frac{\pi}{2}$$

$\therefore$ At $x=-\frac{1}{\sqrt{3}}$

$$L.H.L.=\frac{\pi}{2} \qquad\qquad R.H.L.=-\frac{\pi}{2}$$

$\therefore$ At $x=\frac{1}{\sqrt{3}}$

$$L.H.L.=\frac{\pi}{2} \qquad\qquad R.H.L.=-\frac{\pi}{2}$$

$$\therefore \lim_{x\to\infty}f(x)=\frac{\pi}{2}$$

(iii) Continuity :-

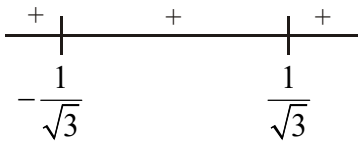
Discontinue at $x=-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}$

(iv) Differentiability :-

$$f'(x)=\begin{cases}\frac{3}{1+x^2} & ; \quad x<-\frac{1}{\sqrt{3}}\\ \frac{3}{1+x^2} & ; \quad -\frac{1}{\sqrt{3}}<x<\frac{1}{\sqrt{3}}\\ \frac{3}{1+x^2} & ; \quad x>\frac{1}{\sqrt{3}}\end{cases}$$

$\therefore$ Not differentiable at $x=\frac{-1}{\sqrt{3}},\frac{1}{\sqrt{3}}$

(v) Monotonocity :-



Always Increasing $\forall x \in R - \left\{-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right\}$

(vi) Concavity

$$f'' = \frac{-6x}{\left(1+x^2\right)^2}; x \in R - \left\{-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right\}$$

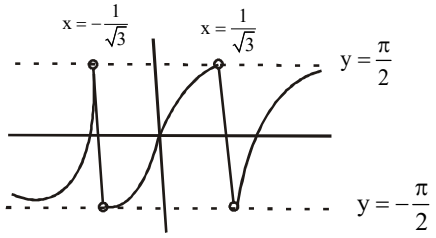
For $x \in \left(-\infty, -\frac{1}{\sqrt{3}}\right)$, $f''(x) > 0$, concave up

$x \in \left(-\frac{1}{\sqrt{3}}, 0\right)$, $f''(x) > 0$, concave up

$x \in \left(0, \frac{1}{\sqrt{3}}\right)$, $f''(x) < 0$ concave down

$x \in \left(\frac{1}{\sqrt{3}}, \infty\right)$, $f''(x) < 0$, concave down

(vii) Graph :-



Note :

1. We can differentiate any identity and the new relation obtained will also be an identity.
2. If $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + ...$

   $f(0) = a_0$ $\quad$ $f'(0) = a_1$
3. If f(x) is divisible by

   (i) By $(x - \alpha)$ $\quad$ then, $f(\alpha) = 0$

   (ii) By $(x - \alpha)^2$ $\quad$ then, $f(\alpha) = 0$, $f'(\alpha) = 0$

   (iii) By $(x - \alpha)^n$; $n \in N$. $\quad$ then, $y(\alpha) = 0$, $y_1(\alpha) = 0$, $y_2(\alpha) = 0$ ..... $y_{n-1}(\alpha) = 0$

5. DIFFERENTIATION OF DETERMINANTS

Let $F(x)=\begin{vmatrix} f & g & h \\ u & v & w \\ l & m & n \end{vmatrix}$ where all functions are differentiable then

$$F'(x)=\begin{vmatrix} f' & g' & h' \\ u & v & w \\ l & m & n \end{vmatrix}+\begin{vmatrix} f & g & h \\ u' & v' & w' \\ l & m & n \end{vmatrix}+\begin{vmatrix} f & g & h \\ u & v & w \\ l' & m' & n' \end{vmatrix}$$

This result may be proved by first principle and the same operation can also be done column wise.

1. $f(x)=\begin{vmatrix} x^2 & x^3 & x^4 \\ 2x & 3x^2 & 4x^3 \\ 1 & 2 & 3 \end{vmatrix}$ find $f'(x)$

Ans. $f'(x)=\begin{vmatrix} x^2 & x^3 & x^4 \\ 2 & 6x & 12x^2 \\ 1 & 2 & 3 \end{vmatrix}$

Sol. $f'(x)=\begin{vmatrix} 2x & 3x^2 & 4x^3 \\ 2x & 3x^2 & 4x^3 \\ 1 & 2 & 3 \end{vmatrix}+\begin{vmatrix} x^2 & x^3 & x^4 \\ 2 & 6x & 12x^2 \\ 1 & 2 & 3 \end{vmatrix}+0$

$$f'(x)=\begin{vmatrix} x^2 & x^3 & x^4 \\ 2 & 6x & 12x^2 \\ 1 & 2 & 3 \end{vmatrix}$$

2. $D=\begin{vmatrix} f & g & h \\ (xf)' & (xg)' & (xh)' \\ (x^2f)'' & (x^2g)'' & (x^2h)'' \end{vmatrix}$ prove that $D'=\begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^3f'')' & (x^3g'')' & (x^3h'')' \end{vmatrix}$

Sol. $x^2f+2x\cdot f+x2f'$

$2x\cdot f'+2f+2x\cdot f'+x'f''$

$$D' = \begin{vmatrix} f & g & h \\ xf'+f & xg'+g & xh'+h \\ x^2f''+4xf'+2f & x^2g''+4xg'+2g & x^2h''+4xh'+2h \end{vmatrix}$$

$\therefore R_3 \to R_3 - 4R_2 + 2R_1$ $\quad\quad$ $\therefore R_2 \to R_2 - R_1$

$$D' = \begin{vmatrix} f & g & h \\ xf'+f & xg'+g & xh'+h \\ x^2f'' & x^2g'' & x^2h'' \end{vmatrix} \qquad D' = \begin{vmatrix} f & g & h \\ xf' & xg' & xh' \\ x^2f'' & x^2g'' & x^2h'' \end{vmatrix}$$

Take x common from R_2 give it to R_3

$$D' = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ x^3f'' & x^3g'' & x^3h'' \end{vmatrix}$$

$\therefore$ Now differentiate w.r.t. x :-

$$D' = \begin{vmatrix} f' & g' & h' \\ f' & g' & h' \\ x^3f'' & x^3g'' & x^3h'' \end{vmatrix} + \begin{vmatrix} f' & g' & h' \\ f'' & g'' & h'' \\ x^3f'' & x^3g'' & x^3h'' \end{vmatrix} + \begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^3f'')' & (x^3g'')' & (x^3h'')' \end{vmatrix}$$

6.1 L'HOSPITAL RULE

This rule is applicable for the indeterminate forms of the type $\frac{0}{0}, \frac{\infty}{\infty}$. If the function f(x) and g(x) are differentiable in certain neighbourhood of the point 'a', except, may be, at the point 'a' itself and $g'(x) \neq 0$, and if

$\lim\limits_{x\to a} f(x) = \lim\limits_{x\to a} g(x) = 0$ or $\lim\limits_{x\to a} f(x) = \lim\limits_{x\to a} g(x) = \infty$,

then $\lim\limits_{x\to a} \frac{f(x)}{g(x)} = \lim\limits_{x\to a} \frac{f'(x)}{g'(x)}$

Note:

1. The point 'a' may be either finite or infinite ($+\infty$ or $-\infty$).

2. f(x) and g(x) are differentiable functions.

3. Applicable only in $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

4. It can be applied multiple times.

5. Do not apply quotient rule.

Q.1 $\lim\limits_{x\to 0}\dfrac{\sin x}{x}=\lim\limits_{x\to 0}\dfrac{\cos x}{1}=1$

Sol. $\lim\limits_{x\to 0}\dfrac{\sin}{x}=\lim\limits_{x\to 0}\dfrac{\cos x}{1}=\dfrac{1}{1}=1$

Q.2 $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x^2}=\lim\limits_{x\to 0}\dfrac{\sin x}{2x}=\lim\limits_{x\to 0}\dfrac{\cos x}{2}=\dfrac{1}{2}$

Sol. $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x^2}=\lim\limits_{x\to 0}\dfrac{0-(-\sin x)}{2x}=\dfrac{1}{2}\lim\limits_{x\to 0}\left[\dfrac{\sin x}{x}\right]=\dfrac{1}{2}$

Q.3 $\lim\limits_{x\to 0}\dfrac{e^x-1}{x}=\dfrac{e^x}{1}=1$

Sol. $\lim\limits_{x\to 0}\dfrac{e^x-1}{x}=\lim\limits_{x\to 0}\dfrac{e^x-0}{1}=1$

Q.4 $\lim\limits_{x\to 0}\dfrac{x-\sin x}{\tan^3 x}=\dfrac{x-\sin x}{x^3}=\dfrac{1-\cos x}{3x^2}=\dfrac{1}{6}$

Sol. $\lim\limits_{x\to 0}\dfrac{x-\sin x}{\tan^3 x}=\lim\limits_{x\to 0}\dfrac{x-\sin x}{\tan^3 x\times\dfrac{x^3}{x^3}}$

$=\lim\limits_{x\to 0}\dfrac{x-\sin x}{x^3}=\lim\limits_{x\to 0}\dfrac{1-\cos x}{3x^2}$

$=\dfrac{1}{3}\times\dfrac{1}{2}=\dfrac{1}{6}$

5. $\lim\limits_{x\to\infty}\dfrac{x^{100}}{e^x}$

Ans. 0

Sol. $\lim\limits_{x\to\infty}\dfrac{x^{100}}{e^x}\Rightarrow$ After 100th derivative of Nr & Dr

$\lim\limits_{x\to\infty}\dfrac{100!}{e^x}$

At $x\to\infty$

$=\dfrac{100!}{e^{\infty}\to\infty}=0$

6. $\lim\limits_{x\to a} \dfrac{x^2 f(a) - a^2 f(x)}{x-a}$

Ans. $2af(a) - a^2 f'(a)$

Sol. $\lim\limits_{x\to a} \dfrac{x^2 f(a) - a^2 f(x)}{x-a}$

$= \lim\limits_{x\to a} \dfrac{2xf(a) - a^2 f'(x)}{1}$

At $x = a$

$2af(a) - a2f'(a)$

7. $\lim\limits_{x\to a} \dfrac{x^x - a^x}{x-a}$

Ans. a^a

Sol. $\lim\limits_{x\to a} \dfrac{x^x - a^x}{x-a} = \lim\limits_{x\to a} \dfrac{x^x(1+\ell nx) - a^x \cdot \ell na}{1}$

$= a^a(1 + \ell na) - a^a \ell na$

$= a^a + a^a \ell na - a^a \ell na$

$= a^a$

8. $\lim\limits_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$

Ans. $\dfrac{1}{2}$

Sol. $\lim\limits_{x\to 0} \dfrac{e^x - 1 - x}{x^2} = \lim\limits_{x\to 0} \dfrac{e^x - 0 - 1}{2x} = \dfrac{1}{2}\lim\limits_{x\to 0} \dfrac{e^x - 1}{x} = \dfrac{1}{2}$

9. $\lim\limits_{x\to 0} \dfrac{x\cos x - \ln(1+x)}{x^2}$

Ans. $\dfrac{1}{2}$

Sol. $= \lim\limits_{x\to 0} \dfrac{x(-\sin x) + \cos x - \dfrac{1}{1+x}}{2x}$

$$= \lim_{x\to 0} \frac{-x\cos x - \sin x - \sin x + \frac{1}{(1+x)^2}}{2} = \frac{1}{2}$$

10. $\lim\limits_{x\to 0} x \ln x$

Ans. 0

Sol. $= \lim\limits_{x\to 0} \frac{\ell n x}{\frac{1}{x}} \Rightarrow \lim\limits_{x\to 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$

At $x = 0$

0

11. $\lim\limits_{x\to 0} \frac{\ell n(\cos x)}{\ell n\left(\sec\frac{x}{2}\right)}$

Ans. -4

Sol. $\lim\limits_{x\to 0} \frac{\ell n(\cos x)}{\ell n\left(\sec\frac{x}{2}\right)}$

$$\lim_{x\to 0} \frac{\left(\frac{-\sin x}{\cos x}\right)}{\frac{\left(\sec\frac{x}{2} + \tan\frac{x}{2}\right)\times\frac{1}{2}}{\sec\frac{x}{2}}}$$

$$\lim_{x\to 0} \frac{-2\tan x}{\tan\frac{x}{2}}$$

$$\lim_{x\to 0}\frac{-2\sec^2 x}{\frac{1}{2}\sec^2\frac{x}{2}}$$

$= -4$

12 $\lim\limits_{x\to 0}\dfrac{1-\cos x\cos 2x\cos 3x\cos 4x}{x^2}$

Ans. 15

Sol. Let $y = \cos x\cos 2x\cos 3x\cos 4x$

$\ell ny = \ell n\cos x + \ell n\cos 2x + \ell n\cos 3x + \ell n\cos 4x$

$$\frac{dy}{dx} = y[-\tan x - 2\tan 2x - 3\tan 3x - 4\tan 4x]$$

$$\lim_{x\to 0}\frac{0-(\cos x\cos 2x\cos 3x\cos 4x)(-\tan x - 2\tan 2x - 3\tan 3x - 4\tan 4x)}{2x}$$

$$= -\frac{1}{2}(-1-4-9-16) = 15$$

13. $\lim\limits_{x\to 0}\log_x\cot x$

Ans. – 1

Sol. $$\lim_{x\to 0}\frac{\ell n\cot x}{\ell nx} \Rightarrow \frac{-\frac{\operatorname{cosec}^2 x}{\cot x}}{\frac{1}{x}}$$

$$= -\lim_{x\to 0} \frac{x\tan x}{\frac{\sin^2 x}{x^2}x^2}$$

$$= -1$$

14. $\lim\limits_{x\to 0} \dfrac{1+\sin x-\cos x+\ln(1-x)}{x^2.\sin x}$

Ans. $-\dfrac{1}{2}$

Sol. $\lim\limits_{x\to 0} \dfrac{1+\sin x-\cos x+\ell n(1-x)}{x^2.\dfrac{\sin x}{x}\times x}$

$$= \lim_{x\to 0} \frac{0+\cos x+\sin x-\frac{1}{1-x}}{3x^2}$$

Again by L′HR :

$$\lim_{x\to 0} \frac{-\sin x+\cos x-\frac{1}{(1-x)^2}}{6x}$$

Again by LHR

$$\lim_{x\to 0} \frac{-\cos x-\sin x-\frac{2}{(1-x)^3}}{6}$$

At x = 0

$$\frac{-1-2}{6} = -\frac{1}{2}$$

6.2 0^0 & ∞^0 FORM

Indeterminate forms of the type ∞^0 or 0^0 are reduced to forms of the type $0\times\infty$ by taking logarithms or by the transformation $[f(x)]^{\phi(x)} = e^{\phi(x).\ell nf(x)}$.

1. $\lim\limits_{x\to 0} x^x$

Ans. 1

Sol $\lim_{x\to 0} x^x = \lim_{x\to 0} e^{x\ln x} = e^{\lim_{x\to 0} x\ln x} = e^L$

$L = \lim_{x\to 0} x\ln x$

$= \text{Lim}_{x\to 0} \frac{\ln x}{1/x} = \text{Lim}_{x\to 0} \frac{1/x}{-1/x^2}$

$= \text{Lim}_{x\to 0} -x$

$= 0$

Ans = $e^L = e^0 = 1$

2. $\lim_{x\to 0} x^{\tan x}$

Ans. 1

Sol. Let $y = x^{\tan x}$

$\ell ny = \tan x\, \ell nx$

Taking limit

$= \lim_{x\to 0} \frac{\ell nx}{\cot x}$

$= \lim_{x\to 0} \frac{\frac{1}{x}}{-\text{cosec}^2 x}$

$= -\lim_{x\to 0} \frac{\sin^2 x}{x^2} \times x$

$\ell ny = 0 \Rightarrow y = e^0 = 1$

3. $\lim_{x\to 0} (\text{cosec } x)^{1/\cot x}$

Ans. 1

Sol. $\ell ny = \lim_{x\to 0} \frac{1}{\cot x} \cdot \ell n \text{ cosec } x$

$= \lim_{x\to 0} \frac{\frac{1}{\text{cosec } x} \times -\text{cosec} x \cdot \cot x}{-\text{cosec}^2 x}$

$$\frac{\sin^2 x}{\frac{\cos x}{\sin x}}$$

At $x = 0$

$$\frac{0}{1} = 0$$

$$y = e^0 = 1$$

4. $\lim\limits_{x\to 1} \left(1-x^2\right)^{\frac{1}{\ln(1-x)}}$

Ans. e^1

Sol. $\ell = \lim\limits_{x\to 1^-} \left(1-x^2\right)^{\frac{1}{\log(1-x)}}$

$$\log \ell = \lim_{x\to 1^-} \frac{1}{\log(1-x)} \log\left(1-x^2\right)$$

$$= \lim_{x\to 1^-} \frac{\ell n(1+x) + \ell n(1-x)}{\ell n(1-x)}$$

$$= \lim_{x\to 1^-} \frac{\ell n(1+x)}{\ell n(1-x)} + 1$$

$$\log l = \frac{\left(\frac{1}{1+x}\right)}{\left(-\frac{1}{1-x}\right)} + 1$$

$$\log \ell = 1$$

$$\ell = e$$

7. DIFFERENTIATION OF INTEGRATION

1. $I = \int f(x) \cdot dx$

$$\frac{dI}{dx} = f(x)$$

2. LEIBNITZ Rule

$$I = \int_{g(x)}^{h(x)} f(t)dt \qquad (f \text{ is continuous })$$

$$\frac{dI}{dx} = f(h(x))\, h'(x) - f(g(x))\, g'(x)$$